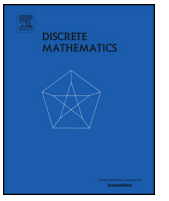
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Bijections in weakly increasing trees via binary trees



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ABSTRACT

As a unification of increasing trees and plane trees, the weakly increasing trees labeled by a multiset was introduced by Lin-Ma-Ma-Zhou in 2021. Motived by some symmetries in plane trees proved recently by Dong, Du, Ji and Zhang, we construct four bijections on weakly increasing trees in the same flavor via switching the role of left child and right child of some specified nodes in their corresponding binary trees. Consequently, bijective proofs of the aforementioned symmetries found by Dong et al. and a non-recursive construction of a bijection on plane trees of Deutsch are provided. Applications of some symmetries in weakly increasing trees to permutation patterns and statistics will also be discussed.

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1. Introduction

Using the approach of grammatical calculus introduced by Chen [3] and further developed by Chen and Fu [5], Dong, Du, Ji and Zhang [11] proved three symmetric joint distributions over plane trees and asked for bijective proofs. Weakly increasing trees introduced by Lin, Ma, Ma and Zhou [19] are common generalization of plane trees and increasing trees. The main objective of this paper is to extend the aforementioned symmetries of Dong et al. from plane trees to weakly increasing trees via three tree involutions constructed in a unified manner.

Recall that a *plane tree* is a rooted tree in which the children of each node are linearly ordered. Throughout this paper, let $M = \{1^{p_1}, 2^{p_2}, \cdots, n^{p_n}\}$ be a multiset with cardinality $|M| = p_1 + \cdots + p_n$.

Definition 1.1 *(Weakly increasing trees [19]).* A *weakly increasing tree* on M is a plane tree with $|M|+1$ nodes that are labeled precisely by all elements in the multiset $M \cup \{0\}$ satisfying

(i) the labels along a path from the root to any leaf are weakly increasing;
(ii) the labels of the children of each node are weakly increasing from right to left.

Denote by $\mathcal{T}_M$ the set of weakly increasing trees on M.

Note that the root of a weakly increasing tree is always labeled by 0. See Fig. 1 (in left) for a weakly increasing tree on $M = \{1^2, 2^4, 3^3, 4^2\}$. It is clear from the above definition that weakly increasing trees on $[n] := \{1, 2, \ldots, n\}$ are exactly *increasing trees* on $[n]$, while weakly increasing trees on $\{1^n\}$ are in obvious bijection with $\mathcal{P}_n$, the set of all plane trees with n

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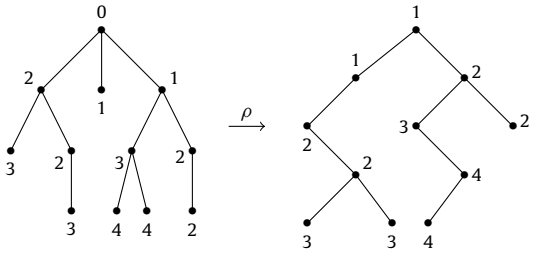


Fig. 1. A weakly increasing tree on $\{1^2, 2^4, 3^3, 4^2\}$ and its corresponding weakly increasing binary tree under the bijection ρ.

edges. In this fashion, one can study the unity between plane trees and increasing trees in the framework of weakly increasing tees. Since the introduction of this notion by Lin et al. [19] in 2021, there has been many interesting developments and unexpected applications found in [6,8,18,20–23], including connections with the Jacobi elliptic functions and combinatorial interpretation of the γ-coefficients of the multiset Eulerian polynomials. Remarkably, it was proved in [19] that

$$|\mathcal{T}_M| = \frac{1}{1+N_n}\prod_{i=1}^{n}\binom{N_i+p_i}{p_i},$$

where $N_i := p_1 + p_2 + \cdots + p_i$ for each $i \in [n]$.

Definition 1.2. In a weakly increasing tree, a node without any child is called a *leaf*, while a node with children is referred to as an *internal node*. Nodes that share the same parent are called *siblings* and those to the left (resp., right) of a node v are termed the *elder* (resp., *younger*) *siblings* of v. A leaf is considered an *old leaf* if it is the leftmost child of its parent, otherwise, it is a *young leaf*. Furthermore, an old leaf is called a *singleton leaf* if it is the only child of its parent, and an *elder leaf* if it has siblings. Finally, an internal node is classified as an *elder internal node* if it is the parent of an elder leaf, a *singleton internal node* if it is the parent of a singleton leaf, and a *young internal node* otherwise.

For a tree $T \in \mathcal{T}_M$, let $\mathsf{sleaf}(T)$ (resp., $\mathsf{eleaf}(T)$, $\mathsf{sint}(T)$, $\mathsf{eint}(T)$, $\mathsf{yleaf}(T)$, $\mathsf{yint}(T)$) denote the number of singleton leaves (resp., elder leaves, singleton internal nodes, elder internal nodes, young leaves, young internal nodes) in T. For example, if T is the first tree of Fig. 1, then all six statistics on T are equal to 2. It is clear that

$$\begin{aligned} &\mathsf{sleaf}(T) = \mathsf{sint}(T), \quad \mathsf{eleaf}(T) = \mathsf{eint}(T) \quad \text{and} \\ &2\mathsf{sleaf}(T) + 2\mathsf{eleaf}(T) + \mathsf{yleaf}(T) + \mathsf{yint}(T) = |M| + 1. \end{aligned} \tag{1.1}$$

Over plane trees, old and young leaves were introduced by Chen, Deutsch and Elizalde [4] in studying some combinatorial identities, while the further refined six statistics were considered only recently by Dong et al. in [11].

Theorem 1.3. *Fix a multiset M with $|M| \geq 2$. There exists an involution Φ on $\mathcal{T}_M$ that transforms the quadruple* (sleaf, eleaf, yleaf, yint) *to* (eleaf, sleaf, yint, yleaf).

For $n \geq 2$, the equidistribution of the two quadruples

$$(\mathsf{sleaf}, \mathsf{eleaf}, \mathsf{yleaf}, \mathsf{yint}) \quad \text{and} \quad (\mathsf{eleaf}, \mathsf{sleaf}, \mathsf{yint}, \mathsf{yleaf})$$

on $\mathcal{P}_n$ was proved algebraically by Dong et al. [11]. Theorem 1.3 extends their equidistribution from plane trees to weakly increasing trees.

Definition 1.4 *(Tip-augmented weakly increasing trees).* A tree in $\mathcal{T}_M$ without any young internal node is called a *tip-augmented weakly increasing tree* on M. The set of all tip-augmented weakly increasing trees on M is denoted by $\mathcal{A}_M$. Note that tip-augmented plane trees were introduced by Donaghey [10].

For a tree $T \in \mathcal{T}_M$, let $\mathsf{oleaf}(T)$ denote the number of old leaves of T and let $\mathsf{oint}(T)$ denote the number of internal nodes which are parents of old leaves of T. Thus,

$$\mathsf{oleaf}(T) = \mathsf{sleaf}(T) + \mathsf{eleaf}(T) \quad \text{and} \quad \mathsf{oint}(T) = \mathsf{sint}(T) + \mathsf{eint}(T).$$

Theorem 1.5. *For any multiset M, we have*

$$\sum_{T\in\mathcal{T}_M} x_1^{\mathsf{oleaf}(T)} x_2^{\mathsf{yleaf}(T)} y_1^{\mathsf{oint}(T)} y_2^{\mathsf{yint}(T)} = \sum_{T\in\mathcal{A}_M} (x_1 y_1)^{\mathsf{oleaf}(T)} (x_2 + y_2)^{\mathsf{yleaf}(T)}. \tag{1.2}$$

For $n \geq 0$, Chen and Pan [7] considered the following two-variable Motzkin polynomials

$$M_n(u, v) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} C_k u^{k+1} v^{n-2k},$$

where $C_n := \frac{1}{1+n}\binom{2n}{n}$ is the n-th *Catalan number*. When u and v are positive integers, the Motzkin polynomial $M_n(u, v)$ was termed the *generalized Motzkin numbers* by Sun [24]. We shall provide a group action proof of Theorem 1.5 and derive from its plane tree case the following relationship between refined Narayana polynomials and Motzkin polynomials.

Corollary 1.6 *(Dong-Du-Ji-Zhang [11]). For $n \geq 1$, we have*

$$\sum_{T \in \mathcal{P}_n} x_1^{\mathsf{oleaf}(T)} x_2^{\mathsf{yleaf}(T)} y_1^{\mathsf{oint}(T)} y_2^{\mathsf{yint}(T)} = M_{n-1}(x_1 y_1, x_2 + y_2). \tag{1.3}$$

Specializing $x_1 = x_2 = x$ and $y_1 = y_2 = y$ in (1.3) leads to a result of Chen and Pan [7, Theorem 1.3 (1.14)]. In order to prove (1.3) using Chen's context-free grammar, Dong et al. [11] considered further refinements of elder leaves and young leaves on plane trees. We extend their notions to weakly increasing trees and introduce new refinements of singleton leaves.

Definition 1.7. In a weakly increasing tree, we define six kinds of leaves:

- a singleton leaf whose parent has elder siblings is called a *singleton leaf with uncle*;
- a singleton leaf whose parent has no elder sibling is called a *singleton leaf with no uncle*;
- an elder leaf is considered as an *elder twin leaf* if the second child of its parent is also a leaf;
- an elder leaf is considered as an *elder non-twin leaf* if the second child of its parent is not a leaf;
- a young leaf is considered as a *second leaf* if it is the second child of its parent;
- a young leaf is considered as a *younger leaf* if it is not the second child of its parent.

For $T \in \mathcal{T}_M$, let $\mathsf{suleaf}(T)$ (resp., $\mathsf{snuleaf}(T)$, $\mathsf{etleaf}(T)$, $\mathsf{entleaf}(T)$, $\mathsf{syleaf}(T)$, $\mathsf{yerleaf}(T)$) denote the number of singleton leaves with uncle (resp., singleton leaves with no uncle, elder twin leaves, elder non-twin leaves, second leaves and younger leaves) in T.

Let $\mathcal{A}_n$ be the set of all tip-augmented plane trees in $\mathcal{P}_n$, i.e., $\mathcal{A}_n = \mathcal{A}_{\{1^n\}}$. Consider a refinement of the Motzkin polynomial $M_n(u, v)$ as

$$M_n(u_1, u_2, u_3; v_1, v_2) := \sum_{T \in \mathcal{A}_{n+1}} u_1^{\mathsf{sleaf}(T)} u_2^{\mathsf{etleaf}(T)} u_3^{\mathsf{entleaf}(T)} v_1^{\mathsf{yerleaf}(T)} v_2^{\mathsf{syleaf}(T)}.$$

Dong et al. [11] proved algebraically the following symmetry on tip-augmented plane trees and asked for a bijective proof.

Proposition 1.8 *(Dong-Du-Ji-Zhang [11]). For $n \geq 2$, we have*

$$M_n(u_1, u_2, u_3; v_1, v_2) = M_n(u_3, u_2, u_1; v_1, v_2). \tag{1.4}$$

We answer their bijective problem by proving the following theorem; we will later see that it generalizes Proposition 1.8.

Theorem 1.9. *Fix a multiset M with $|M| \geq 2$. There exists an involution Ψ on $\mathcal{T}_M$ that preserves the quadruple* (snuleaf, etleaf, syleaf, yerleaf) *but exchanges the pair* (suleaf, entleaf). *Moreover, Ψ preserves the statistic* yint *and so Ψ restricts to an involution on $\mathcal{A}_M$ which proves the symmetry* (1.4).

Remark 1.10. A different involution proof of Proposition 1.8 via Chen's general bijective algorithm for trees [2] was given by Yang and Zhang [27] very recently.

It turns out that the proofs of Theorems 1.3, 1.5 and 1.9 are all based on a simple switching operation on binary trees, namely switching left and right branches of a node. Moreover, this operation enables us to build non-recursively a bijection due to Deutsch [9] on plane trees and independently extended to weakly increasing trees by Lin–Ma [22] and Chen–Fu [6]. As by a product, we find that the statistic of "internal nodes on even-levels" has the same distribution as "oleaf" over plane trees (see Corollary 2.18). This paper reflects the principle: whenever you encounter constructing bijections on weakly increasing trees, then expect that binary trees will facilitate your problem.

Applications of Theorems 1.3, 1.5 and 1.9 to permutation patterns and statistics will also be discussed, including a generalization of a symmetry regarding "maximal number of non-overlapping descents/ascents" on 231-avoiding permutations due to Kitaev and Zhang [16].

Table 1
Statistics for a tree $T \in \mathcal{T}_M$ that appear in the introduction. See Definitions 1.2 and 1.7 for the terminology on trees.

sleaf(T)	#{singleton leaves in T}
eleaf(T)	#{elder leaves in T}
sint(T)	#{singleton internal nodes in T}
eint(T)	#{elder internal nodes in T}
yleaf(T)	#{young leaves in T}
yint(T)	#{young internal nodes in T}
oleaf(T)	#{old leaves in T}
oint(T)	#{parents of old leaves in T}
suleaf(T)	#{singleton leaves with uncle in T}
snuleaf(T)	#{singleton leaves with no uncle in T}
etleaf(T)	#{elder twin leaves in T}
entleaf(T)	#{elder non-twin leaves in T}
syleaf(T)	#{second leaves in T}
yerleaf(T)	#{younger leaves in T}

The rest of this paper is organized as follows. In Section 2, we prove Theorems 1.3, 1.5 and 1.9 and provide a non-recursive construction of Deutsch's bijection using a simple switching operation on binary trees. Using binary trees, we can also prove a unified formula that generalizes two main formulas of Dong et al. [11]. Section 3 is devoted to some applications of our bijections to permutation patterns and statistics, including two symmetries in 312-avoiding permutations, one of which generalizes a result of Kitaev and Zhang [16].

For the reader's convenience, we list in Table 1 all the statistics on weakly increasing trees appearing in the introduction.

2. Combinatorics of weakly increasing trees via binary trees

In this section, via the switching operation on binary trees, we prove Theorems 1.3, 1.5 and 1.9 and provide a non-recursive construction of a bijection on plane trees due to Deutsch. The generating function for the quintuple (sleaf, etleaf, entleaf, syleaf, yerleaf) over plane trees will also be computed via binary trees, unifying two generating function formulas due to Dong et al. [11].

2.1. Weakly increasing binary trees

We begin by introducing weakly increasing binary trees, which are in natural bijection with weakly increasing trees. A *binary tree* is a rooted tree where each internal node has either a left child, a right child, or both. Let $\mathcal{B}_n$ denote the set of all binary trees with n nodes. As a common generalization of binary trees and binary increasing trees, the weakly increasing binary trees was first used in [22].

Definition 2.1 *(Weakly increasing binary trees [22]).* For a fixed multiset M, a *weakly increasing binary tree* on M is a labeled binary tree such that

(i) the labels of the nodes form precisely the multiset M and
(ii) the labels along a path from the root to any leaf are weakly increasing.

Denote by $\mathcal{B}_M$ the set of weakly increasing binary trees on M.

See Fig. 1 (in right) for a weakly increasing binary tree on $M = \{1^2, 2^4, 3^3, 4^2\}$. Note that weakly increasing binary trees on $[n]$ are precisely increasing binary trees on $[n]$, while weakly increasing binary trees on $\{1^n\}$ are in bijection with $\mathcal{B}_n$. For a weakly increasing tree $T \in \mathcal{T}_M$, we define its corresponding weakly increasing binary tree $\rho(T) \in \mathcal{B}_M$ by imposing the condition on each pair of non-root nodes (x, y) in T:

(i) y is the right child of x in $\rho(T)$ only when y is the rightmost child of x in T;
(ii) y is the left child of x in $\rho(T)$ only when y is the closest elder sibling of x in T.

The mapping $T \mapsto \rho(T)$ establishes a natural one-to-one correspondence between $\mathcal{T}_M$ and $\mathcal{B}_M$, which extends the classical bijection between $\mathcal{P}_n$ and $\mathcal{B}_n$ in [26, Page 9]. See Fig. 1 for an example of the bijection ρ. In Table 2, we consider all possibilities of the status of children of the node i in a binary tree $B \in \mathcal{B}_M$ and write the corresponding status of the node i in the tree $\rho^{-1}(B) \in \mathcal{T}_M$.

For a binary tree $B \in \mathcal{B}_M$ and a node x of B, let $\phi_x(B)$ be the tree in $\mathcal{B}_M$ obtained from B by switching the left branch and the right branch of x, if any. For any two nodes x, y of B, it is clear that $\phi_x \circ \phi_y(T) = \phi_y \circ \phi_x(T)$, i.e., ϕ_x and ϕ_y are commute. Thus, if A is a set of some specified nodes of B, then it is reasonable to define the transformation $B \mapsto \left(\prod_{x \in A} \phi_x\right)(B)$.

Table 2
Correspondence between status of each node i in a binary tree $B \in \mathcal{B}_M$ and the node i in $\rho^{-1}(B) \in \mathcal{T}_M$.

Status of node i in binary tree $B \in \mathcal{B}_M$	Status of node i in tree $\rho^{-1}(B) \in \mathcal{T}_M$
has only right child	internal node with no elder sibling
has only left child	young leaf
has both left and right children	internal node with elder sibling
has no children	old leaf

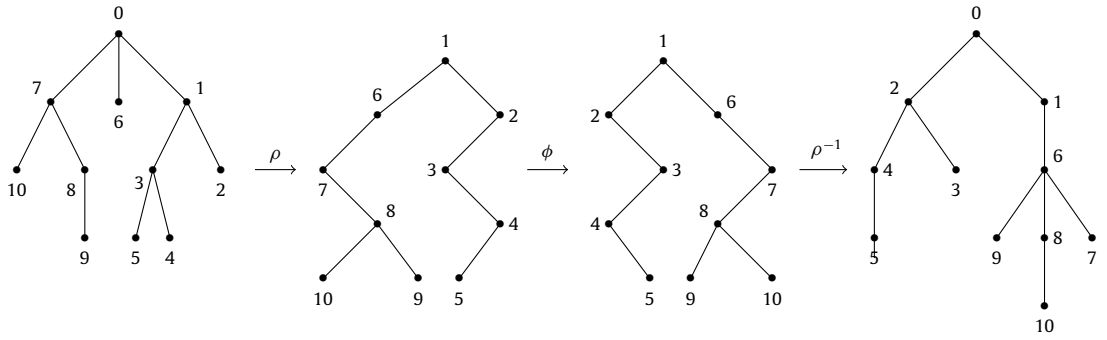


Fig. 2. An example of the involution $\Phi = \rho^{-1} \circ \phi \circ \rho$.

Throughout this paper, if st is a statistic on $\mathcal{T}_M$ and B is a binary tree in $\mathcal{B}_M$, then we define

$$\mathsf{st}(B) = \mathsf{st}(\rho^{-1}(B)),$$

for the sake of convenience. For example, if $B \in \mathcal{B}_M$ with $|M| \geq 2$, then $\mathsf{sleaf}(B)$ is the number of right leaves of B by Lemma 2.2 (i).

2.2. Proofs of Theorems 1.3 and 1.9

The proof of Theorem 1.3 is based on the following key observation.

Lemma 2.2. *Let T be a weakly increasing tree with at least two edges. Then,*

(i) $\mathsf{sleaf}(T)$ *equals the number of right leaves in* $\rho(T)$*;*
(ii) $\mathsf{eleaf}(T)$ *equals the number of left leaves in* $\rho(T)$*;*
(iii) $\mathsf{yleaf}(T)$ *equals the number of nodes having only left child in* $\rho(T)$*;*
(iv) $\mathsf{yint}(T)$ *equals the number of nodes having only right child in* $\rho(T)$*.*

Proof. We will only show (iv), as the other three assertions follow easily from the construction of ρ. Note that a node is a young internal node if and only if its eldest child is an internal node. Since the map ρ assigns an internal eldest node of T to a node in $\rho(T)$ that lacks a left child but possesses right child, (iv) follows. □

Proof of Theorem 1.3. For $B \in \mathcal{B}_M$, let $\phi(B)$ be the minor symmetry of T, i.e., $\phi(B) = \left(\prod_x \phi_x\right)(B)$, running over all nodes x of B. Then the composition $\Phi = \rho^{-1} \circ \phi \circ \rho$ is an involution with the desired property in view of Lemma 2.2. See Fig. 2 for an example of the involution Φ. □

Remark 2.3. A slight modification of the involution Φ was used in [18] to study other symmetry of trees.

Lemma 2.4. *Let T be a weakly increasing tree with at least two edges. Then,*

(i) $\mathsf{suleaf}(T)$ *equals the number of right leaves in* $\rho(T)$ *whose parent has left child;*
(ii) $\mathsf{snuleaf}(T)$ *equals the number of right leaves in* $\rho(T)$ *whose parent has no left child;*
(iii) $\mathsf{etleaf}(T)$ *equals the number of left leaves in* $\rho(T)$ *whose parent has no right child;*
(iv) $\mathsf{entleaf}(T)$ *equals the number of left leaves in* $\rho(T)$ *whose parent has right child;*
(v) $\mathsf{yerleaf}(T)$ *equals the number of nodes in* $\rho(T)$ *with only left child and whose left child has left child;*
(vi) $\mathsf{syleaf}(T)$ *equals the number of nodes in* $\rho(T)$ *with only left child and whose left child has no left child.*

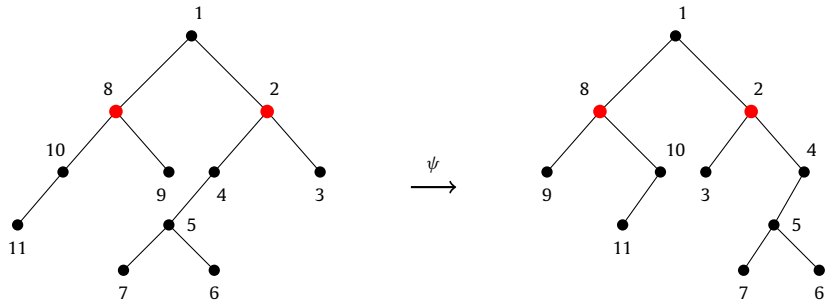


Fig. 3. An example of the involution ψ.

Proof. All the assertions can be verified routinely from the construction of ρ. We leave the details to the reader. □

Proof of Theorem 1.9. For a binary tree $B \in \mathcal{B}_M$ and x a node of B, then x is said to be *unbalanced* if one of its child is a leaf but the other child is an internal node. For example, the unbalanced nodes of the binary tree in Fig. 3 (in left) are 2 and 8. Let $\mathrm{Unb}(B)$ be the set of all unbalanced nodes of B. Let

$$\psi(B) := \left(\prod_{x\in \mathrm{Unb}(T)} \phi_x\right)(B).$$

It is plain to see that $\mathrm{Unb}(B) = \mathrm{Unb}(\psi(B))$ and so ψ is an involution on $\mathcal{B}_M$. See Fig. 3 for an example of ψ, where unbalanced nodes are marked in red. Set $\Psi = \rho^{-1} \circ \psi \circ \rho$. As ψ is an involution on $\mathcal{B}_M$, Ψ is an involution on $\mathcal{T}_M$.

In order to prove the first statement, we need to show that

(a) ψ preserves the quadruple $(\mathsf{snuleaf}, \mathsf{etleaf}, \mathsf{syleaf}, \mathsf{yerleaf})$;
(b) for any $B \in \mathcal{B}_M$, $\mathsf{suleaf}(B) = \mathsf{entleaf}(\psi(B))$.

Here the six statistics $(\mathsf{snuleaf}, \mathsf{etleaf}, \mathsf{syleaf}, \mathsf{yerleaf}, \mathsf{suleaf}, \mathsf{entleaf})$ over binary trees have their meanings as proved in Lemma 2.4. Assertion (a) follows from the fact that

$$(\mathsf{snuleaf}, \mathsf{etleaf}, \mathsf{syleaf}, \mathsf{yerleaf})B = (\mathsf{snuleaf}, \mathsf{etleaf}, \mathsf{syleaf}, \mathsf{yerleaf})\phi_x(B)$$

whenever x is an unbalanced node of B. To see assertion (b), we note that

$$\mathsf{suleaf}(B) = |\{x \in \mathrm{Unb}(B) : \text{ right child of } x \text{ is a leaf}\}| + \mathsf{twin}(B),$$
$$\mathsf{entleaf}(B) = |\{x \in \mathrm{Unb}(B) : \text{ left child of } x \text{ is a leaf}\}| + \mathsf{twin}(B),$$

where $\mathsf{twin}(B)$ is the number of nodes of B whose two children are leaves. Assertion (b) now follows from the fact that if x is an unbalanced node whose right child is a leaf in B, then x becomes an unbalanced node whose left child is a leaf in $\psi(B)$. This completes the proof of the first statement.

By Lemma 2.2 (iv), $\mathsf{yint}(B)$ is the number of nodes having only right child in B for any $B \in \mathcal{B}_M$. Thus, ψ preserves the statistic yint. Moreover, since

$$\mathsf{sleaf}(T) = \mathsf{snuleaf}(T) + \mathsf{suleaf}(T)$$

and that in a tip-augmented plane tree $\mathsf{snuleaf}(T) = 0$, Proposition 1.8 is a special case of Theorem 1.9. This completes the proof of the theorem. □

2.3. A group action proof of Theorem 1.5 and a related bijective problem of Dong et al.

We use the preorder [26, Page 10] (i.e., recursively traversing the parent to the left subtree then to the right subtree) to order the nodes of binary trees. Fix a multiset M with $|M| = m$. For $B \in \mathcal{B}_M$ and $i \in [m]$, suppose that x is the i-th (under preorder) node of B and define

$$\varphi_i(B) = \begin{cases} \phi_x(B), & \text{if } x \text{ has exactly one child in } B; \\ B, & \text{otherwise.} \end{cases}$$

It is clear that the order of the nodes in B remains the same as in $\varphi_i(B)$ and so $\varphi_i^2(B) = B$. As φ_i and φ_j commute for all $i, j \in [m]$, for any subset $S \subseteq [m]$, we can define the function $\varphi_S : \mathcal{B}_M \to \mathcal{B}_M$ by

$$\varphi_S(B) = \left(\prod_{i\in S}\varphi_i\right)(B).$$

The group $\mathbb{Z}_2^m$ acts on $\mathcal{B}_M$ via the functions φ_S, $S \subseteq [m]$. By Lemma 2.2, we have

$$\sum_{T\in\mathcal{T}_M} x_1^{\mathrm{oleaf}(T)} x_2^{\mathrm{yleaf}(T)} y_1^{\mathrm{oint}(T)} y_2^{\mathrm{yint}(T)} = \sum_{B\in\mathcal{B}_M} (x_1y_1)^{\mathrm{oleaf}(B)} x_2^{\mathrm{yleaf}(B)} y_2^{\mathrm{yint}(B)}, \tag{2.1}$$

where $\mathrm{oleaf}(B)$ is the number of leaves in B, $\mathrm{yleaf}(B)$ (resp., $\mathrm{yint}(B)$) is the number of nodes having only left (resp., right) child in B. The following property of φ_i is obvious from its definition.

Lemma 2.5. *For $B \in \mathcal{B}_M$ and $i \in [m]$,*

(i) $\mathrm{oleaf}(B) = \mathrm{oleaf}(\varphi_i(B))$;
(ii) *the i-th node of B has only left (resp., right) child if and only if the i-th node of $\varphi_i(B)$ has only right (resp., left) child.*

We are ready for the proof of Theorem 1.5.

Proof of Theorem 1.5. For any $B \in \mathcal{B}_M$, let $[B] := \{g(B) : g \in \mathbb{Z}_2^m\}$ be the orbit of B under the $\mathbb{Z}_2^m$-action. By Lemma 2.5, each orbit $[B]$ contains a unique binary tree $\tilde{B}$ with $\mathrm{yint}(\tilde{B}) = 0$. Thus, by Lemma 2.5 we have

$$\sum_{B\in[\tilde{B}]} (x_1y_1)^{\mathrm{oleaf}(B)} x_2^{\mathrm{yleaf}(B)} y_2^{\mathrm{yint}(B)} = (x_1y_1)^{\mathrm{oleaf}(\tilde{B})}(x_2+y_2)^{\mathrm{yleaf}(\tilde{B})}$$

for any $\tilde{B} \in \mathcal{B}_M$ with $\mathrm{yint}(\tilde{B}) = 0$. Summing over all orbits yields

$$\sum_{B\in\mathcal{B}_M} (x_1y_1)^{\mathrm{oleaf}(B)} x_2^{\mathrm{yleaf}(B)} y_2^{\mathrm{yint}(B)} = \sum_{\substack{\tilde{B}\in\mathcal{B}_M \\ \mathrm{yint}(\tilde{B})=0}} (x_1y_1)^{\mathrm{oleaf}(\tilde{B})}(x_2+y_2)^{\mathrm{yleaf}(\tilde{B})}.$$

Combining with (2.1) gives (1.2). □

Remark 2.6. Under the bijection ρ, the $\mathbb{Z}_2^m$ group action used above is essentially the same as the generalized Foata–Strehl action on $\mathcal{T}_M$ introduced in [19, Section 7].

Let

$$\mathcal{P}_e(M,k) := \{T \in \mathcal{T}_M : \mathrm{oleaf}(T) = k, \mathrm{yleaf}(T) \text{ is even}\}$$

and

$$\mathcal{P}_o(M,k) := \{T \in \mathcal{T}_M : \mathrm{oleaf}(T) = k, \mathrm{yleaf}(T) \text{ is odd}\}.$$

As

$$2\mathrm{oleaf}(T) + \mathrm{yleaf}(T) + \mathrm{yint}(T) = |M| + 1 \tag{2.2}$$

for any $T \in \mathcal{T}_M$, Setting $x_2 = -1$ and $y_1 = y_2 = 1$ in (1.2) results in the following combinatorial assertion.

Corollary 2.7. *Fix a multiset M with $|M| = m$. Then, for $1 \le k \le \lfloor \frac{m}{2} \rfloor$:*

$$|\mathcal{P}_e(M,k)| = |\mathcal{P}_o(M,k)|. \tag{2.3}$$

Remark 2.8. Corollary 2.7 for plane trees (i.e., $M = \{1^m\}$) was derived from (1.3) by Dong et al. [11], who also asked for a bijective proof. A bijective proof of (2.3) can be easily constructed as follows. By (2.2) and the condition $1 \le k \le \lfloor \frac{m}{2} \rfloor$, we deduce that $\mathrm{yleaf}(T) + \mathrm{yint}(T) > 0$ for any $T \in \mathcal{P}_e(M,k)$. Given $T \in \mathcal{P}_e(M,k)$, find the first node (under preorder), say x, having only one child in $\rho(T)$ and then set $T' = \rho^{-1}\phi_x(\rho(T))$. The map $T \mapsto T'$ sets up an one-to-one correspondence between $\mathcal{P}_e(M,k)$ and $\mathcal{P}_o(M,k)$.

Proof of Corollary 1.6. By the plane tree case of Theorem 1.5, all we need to prove is

$$\sum_{T\in\mathcal{A}_{n+1}} u^{\mathrm{oleaf}(T)} v^{\mathrm{yleaf}(T)} = \sum_{k=0}^{\lfloor n/2\rfloor} \binom{n}{2k} C_k u^{k+1} v^{n-2k}. \tag{2.4}$$

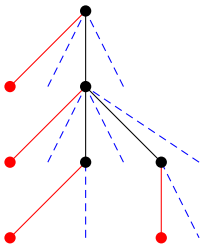


Fig. 4. An example of constructing a tip-augmented plane tree.

To form a tip-augmented plane tree with $k+1$ old leaves, we can first choose a plane trees with k edges, append one node, as old leaf, to each of its $k+1$ nodes, and finally insert $n-2k$ nodes, as young leaves, immediately to the right of some of the $2k+1$ edges (allowing for repetition). See the illustration in Fig. 4 for $k=3$. There are C_k ways to choose a plane tree and $\binom{\binom{2k+1}{n-2k}}{} = \binom{(2k+1)+(n-2k)-1}{n-2k} = \binom{n}{2k}$ choices for inserting the $n-2k$ nodes in the available $2k+1$ positions in the final step, which proves (2.4), as desired. □

It is well known (see [2]) that the number of plane trees with n edges and k leaves is the *Narayana number* $N_{n,k} = \frac{1}{n}\binom{n}{k}\binom{n}{k-1}$. One consequence of Corollary 1.6 is the following explicit formula for the refined Narayana numbers

$$N_{n,k,l} := |\{T \in \mathcal{P}_n : \mathsf{oleaf}(T) = k, \mathsf{yleaf}(T) = l\}|$$

due to Chen, Deutsch and Elizalde [4].

Corollary 2.9 *(Chen-Deutsch-Elizalde [4]). For $n, k \geq 1$ and $l \geq 0$, we have*

$$N_{n,k,l} = \frac{1}{n}\binom{n}{k}\binom{n-k}{l}\binom{n-k-l}{k-1}.$$

Proof. Setting $y_1 = y_2 = 1$ in (1.3) gives

$$\sum_{T\in\mathcal{P}_n} x_1^{\mathsf{oleaf}(T)} x_2^{\mathsf{yleaf}} = \sum_{k=0}^{\lfloor (n-1)/2\rfloor} \binom{n-1}{2k} C_k x_1^{k+1}(1+x_2)^{n-1-2k}.$$

Extracting the coefficient of $x_1^k x_2^l$ yields

$$N_{n,k,l} = \binom{n-1}{2k-2} C_{k-1} \binom{n+1-2k}{l} = \frac{1}{n}\binom{n}{k}\binom{n-k}{l}\binom{n-k-l}{k-1},$$

as desired. □

2.4. *Generating function for* (sleaf, etleaf, entleaf, yerleaf, syleaf) *on plane trees*

Consider the refined Narayana polynomials

$$G_n(x_1, x_2, x_3, x_4) := \sum_{T\in\mathcal{P}_n} x_1^{\mathsf{sleaf}(T)} x_2^{\mathsf{eleaf}(T)} x_3^{\mathsf{yleaf}(T)} x_4^{\mathsf{yint}(T)} \quad \text{for } n \geq 2$$

and $G_1(x_1, x_2, x_3, x_4) = x_2$. The main results of [11] are two formulas, derived from Chen's context-free grammar, for the generating functions of $G_n(x_1, x_2, x_3, x_4)$ and the refined Motzkin polynomials $M_n(u_1, u_2, u_3; v_1, v_2)$ defined in the introduction. Using binary trees, we can prove a unified formula that generalizes their two formulas.

We consider the following new refined Narayana polynomials.

Definition 2.10 *(New refined Narayana polynomials).* Let

$$N_n = N_n(u_1, u_2, u_3; v_1, v_2) := \sum_{T\in\mathcal{P}_n} u_1^{\mathsf{sleaf}(T)} u_2^{\mathsf{etleaf}(T)} u_3^{\mathsf{entleaf}(T)} v_1^{\mathsf{yerleaf}(T)} v_2^{\mathsf{syleaf}(T)}$$

for $n \geq 2$ and $N_1 = N_1(u_1, u_2, u_3; v_1, v_2) := u_1$ by convention.

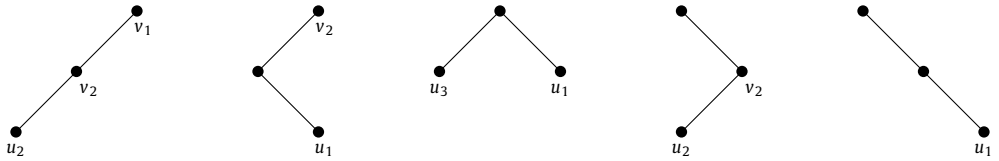


Fig. 5. 3-node binary trees.

The polynomials N_n for $2 \le n \le 5$ are listed as follows:

$$\begin{aligned}
N_2 =& u_1 + u_2v_2,\\
N_3 =& u_1 + u_1u_3 + u_1v_2 + u_2v_2 + u_2v_1v_2,\\
N_4 =& u_2v_1^2v_2 + u_1u_2v_2 + u_1u_3v_1 + u_1v_1v_2 + u_2u_3v_2 + u_2v_1v_2 + u_2v_2^2\\
&+ u_1^2 + 2u_1u_3 + 2u_1v_2 + u_2v_2 + u_1,\\
N_5 =& u_2v_1^3v_2 + 2u_1u_2v_1v_2 + u_1u_3v_1^2 + u_1v_1^2v_2 + u_2^2v_2^2 + 2u_2u_3v_1v_2 + u_2v_1^2v_2\\
&+ 2u_2v_1v_2^2 + u_1^2u_3 + u_1^2v_1 + u_1^2v_2 + 4u_1u_2v_2 + u_1u_3^2 + 2u_1u_3v_1 + 2u_1u_3v_2\\
&+ 2u_1v_1v_5 + u_1v_2^2 + 2u_2u_3v_2 + u_2v_1v_2 + 2u_2v_2 + 3u_1^2 + 3u_1u_3 + 3u_1v_2\\
&+ u_2v_2 + u_1.
\end{aligned}$$

By Lemma 2.4, we have

$$N_n = \sum_{B\in\mathcal{B}_n} u_1^{\mathsf{sleaf}(B)}u_2^{\mathsf{etleaf}(B)}u_3^{\mathsf{entleaf}(B)}v_1^{\mathsf{yerleaf}(B)}v_2^{\mathsf{syleaf}(B)},$$

where

- $\mathsf{sleaf}(B)$ is the number of right leaves of B;
- $\mathsf{etleaf}(B)$ is the number of left leaves of B whose parent has no right child;
- $\mathsf{entleaf}(B)$ is the number of left leaves of B whose parent has right child;
- $\mathsf{yerleaf}(B)$ is the number of nodes of B with only left child and whose left child has left child;
- $\mathsf{syleaf}(B)$ is the number of nodes of B with only left child and whose left child has no left child.

Fig. 5 provides all binary trees in $\mathcal{B}_3$, where the statistics are marked by their corresponding variables.

Theorem 2.11. *The expression of $N(t) := \sum_{n\ge1} N_nt^n$ is given by*

$$N(t) = \frac{at^2 - ct + 1}{2t} - \frac{\sqrt{a^2t^4 + 2(bc - 2u_1 - 2u_2v_2)t^3 + (c^2 - 2b)t^2 - 2ct + 1}}{2t}, \tag{2.5}$$

where $a = u_1 - u_3 + v_1 - v_2$, $b = u_1 + u_3 - v_1 + v_2$ and $c = v_1 + 1$.

Proof. We derive the functional equation for $N(t)$ using the classical decomposition of binary trees $B = (B_g, r, B_d)$, where r is the root of B and B_g, B_d are respectively the left and right branches at r. We distinguish four cases below.

(1) $B_g = B_d = \emptyset$. This case contributes u_1t to $N(t)$.
(2) $B_g = \emptyset$ and $B_d \ne \emptyset$. This case contributes $tN(t)$.
(3) $B_g \ne \emptyset$ and $B_d \ne \emptyset$. This case contributes $t(N(t) + u_3t - u_1t)N(t)$, as when B_g contains only one node, then such node should contribute u_3t other than u_1t.
(4) $B_g \ne \emptyset$ and $B_d = \emptyset$. We further distinguish three subcases.
   (4a) B_g contains only one node. This case contributes $u_2v_2t^2$.
   (4b) The root of B_g has only right child (but no left child). This case contributes $v_2t^2N(t)$.
   (4c) B_g is not in cases (4a) and (4b). This case contributes $v_1t(N(t) - u_1t - tN(t))$.

Summing over all the contributions from the above cases yields the functional equation for $N = N(t)$:

$$N = t(u_1 + N + (N + u_3t - u_1t)N + u_2v_2t + v_2tN + v_1(N - u_1t - tN)). \tag{2.6}$$

Solving this equation gives (2.5). □

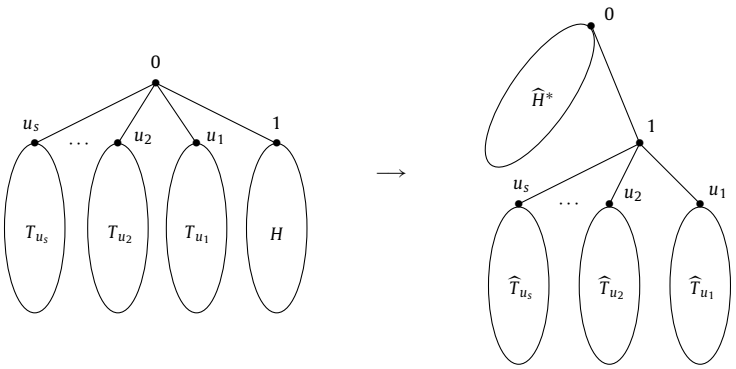


Fig. 6. The construction of $\widehat{T}$.

Remark 2.12. By (2.2), we have

$$\mathsf{yint}(T) = n + 1 - 2\mathsf{oleaf}(T) - \mathsf{yleaf}(T)$$

for $T \in \mathcal{P}_n$. Thus, $N_n(u_1, u_2, u_3; v_1, v_2)$ are common generalization of $G_n(x_1, x_2, x_3, x_4)$ and $M_n(u_1, u_2, u_3; v_1, v_2)$. The main results in [11, Theorems 1.6 and 1.9] can then be deduced from Theorem 2.11 after appropriate substitutions.

2.5. Deutsch's bijection via binary trees

For $T \in \mathcal{T}_M$, the *level* of a node in T is the number of edges on the path from the root to it. The *degree* of a node is the number of its children. Six concerned statistics on T are:

- The number of nodes of degree q in T, denoted by $\mathsf{deg}_q(T)$. In particular, $\mathsf{deg}_0(T) = \mathsf{leaf}(T)$, the number of leaves of T.
- The number of nodes of degree q in odd levels of T, denoted by $\mathsf{od}_q(T)$.
- The number of nodes in even levels of T, denoted by $\mathsf{el}(T)$.
- The number of internal nodes in even levels of T, denoted by $\mathsf{elint}(T)$.
- The number of leaves in even levels of T, denoted by $\mathsf{elleaf}(T)$.
- A non-root node of T is considered as *youngest* if it is the rightmost child of its parent. The number of youngest leaves of T is denoted by $\mathsf{ystleaf}(T)$.

Inspired by the work of Kuznetsov, Pak and Postnikov [17], Deutsch [9] introduced a recursive bijection $\widehat{()} : \mathcal{P}_n \to \mathcal{P}_n$ such that for each $T \in \mathcal{P}_n$:

$$\mathsf{deg}_q(T) = \begin{cases} \mathsf{od}_{q-1}(\widehat{T}) & \text{if} \quad q \geq 1, \\ \mathsf{el}(\widehat{T}) & \text{if} \quad q = 0. \end{cases}$$

This bijection was extended independently by Lin–Ma [22] and Chen–Fu [6] from plane trees to weakly increasing trees. For the sake of convenience, we recall the recursive construction of $\widehat{()} : \mathcal{T}_M \to \mathcal{T}_M$ from [22].

The map $\widehat{()} : \mathcal{T}_M \to \mathcal{T}_M$

Firstly, set $\widehat{\emptyset} = \emptyset$. Let $T \in \mathcal{T}_M$. For a node v of T let T_v denote the subtree of T rooted at v. Clearly, the rightmost child of the root 0 in T must be a node with label 1 and let H denote the subtree of T rooted at this 1. Suppose that all children of the root 0 other than the rightmost child in T from right to left are $u_1, \ldots, u_s$. We construct recursively the weakly increasing tree $\widehat{T}$ as follows:

- Change the label 1 of the root of $\widehat{H}$ to 0 and denote by $\widehat{H}^*$ the resulting tree;
- Attach an isolate node 1 to the root 0 of $\widehat{H}^*$ as its rightmost child;
- Make $\widehat{T}_{u_1}, \ldots, \widehat{T}_{u_s}$ the branches of the above new node 1 from right to left.

See Fig. 6 for the visualization of $\widehat{()}$.

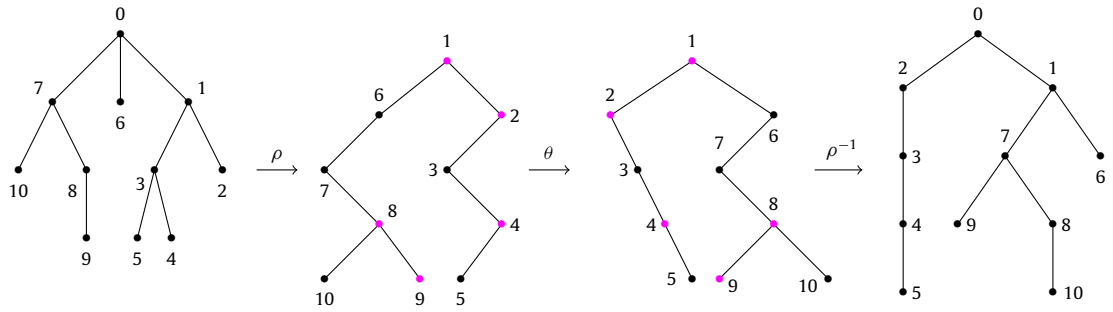


Fig. 7. An example of the bijection $\Theta = \rho^{-1} \circ \theta \circ \rho$.

The bijection $\widehat{()} : \mathcal{T}_M \to \mathcal{T}_M$ can be constructed directly using binary trees as follows. Let $B \in \mathcal{B}_M$ be a binary tree. A node is called a *head* in B if either it is the root or it is a right child in B. Let $H(B)$ be the set of all heads in B. For instance, if B is the second tree in Fig. 7, then $H(B) = \{1, 2, 4, 8, 9\}$. Define the map $\theta : \mathcal{B}_M \to \mathcal{B}_M$ by

$$\theta(B) = \Big(\prod_{x \in H(B)} \phi_x \Big)(B).$$

See Fig. 7 for an example of θ.

Theorem 2.13. *Set $\Theta = \rho^{-1} \circ \theta \circ \rho$. For any $T \in \mathcal{T}_M$, $\Theta(T) = \widehat{T}$.*

Proof. This can be proved by induction on the number of nodes of T, which is straightforward and will be left to the interested reader. □

For a tree $B \in \mathcal{B}_M$ and a node v of B, if there are $i-1$ right edges lying on the path from the root to v, then v is said to be at *right-level* i. Let $\mathrm{Orl}(B)$ be the set of nodes at odd right-levels in B. For example, $\mathrm{Orl}(B) = \{1, 2, 4, 8, 9\}$ if B is the third tree in Fig. 7.

Lemma 2.14. *For any $B \in \mathcal{B}_M$, $H(B) = \mathrm{Orl}(\theta(B))$.*

Proof. We aim to prove that for any node v,

$$(\star) \qquad v \text{ is in } H(B) \Leftrightarrow v \text{ is in } \mathrm{Orl}(\theta(B))$$

by induction on the *depth* of v, that is the number of edges on the path from the root to v.

Clearly, $(\star)$ is true when v is the root, the only node with depth 0. Suppose that $v \in H(B)$ is a non-root node and let y be its parent in B. If $y \in H(B)$ (resp., $y \notin H(B)$), then v is the left (resp., right) child of y in $\theta(B)$. As y has depth strictly less than that of v, then by induction $y \in \mathrm{Orl}(\theta(B))$ (resp. $y \notin \mathrm{Orl}(\theta(B))$), which implies $v \in \mathrm{Orl}(\theta(B))$. The fact that $v \notin H(B)$ implies $v \notin \mathrm{Orl}(\theta(B))$ follows similarly. This proves $(\star)$ for all nodes v by induction. □

By Lemma 2.14, the inverse of θ can be defined as

$$\theta^{-1}(B) = \Big(\prod_{x \in \mathrm{Orl}(B)} \phi_x \Big)(B)$$

for $B \in \mathcal{B}_M$. Thus, the inverse of Θ is $\Theta^{-1} = \rho^{-1} \circ \theta^{-1} \circ \rho$. This direct construction of Deutsch's bijection $\Theta = \widehat{()}$ enables us to observe its further features.

Let $T \in \mathcal{T}_M$ and let v be a node of T. We associate v with a node $\tilde{v}$ as follows:

(i) If v is an internal node, then $\tilde{v}$ is the youngest child of v;
(ii) If v is a leaf that is not youngest, then set $\tilde{v} = v$;
(iii) If v is a youngest leaf, then $\tilde{v}$ is the first node that is not youngest in the path from v to the root.

For the first tree in Fig. 7, we have

$$\tilde{0} = 1, \tilde{1} = 2, \tilde{2} = 0, \tilde{3} = 4, \tilde{4} = 3, \tilde{5} = 5, \tilde{6} = 6, \tilde{7} = 8, \tilde{8} = 9, \tilde{9} = 7, \widetilde{10} = 10.$$

It is plain to see that $v \mapsto \tilde{v}$ is a bijection on all nodes of T.

Theorem 2.15. *For $T \in \mathcal{T}_M$ and v a node of T, then the following assertions hold.*

(i) *If v is an internal node in T with degree q, then $\tilde{v}$ is an odd-level node with degree $q-1$ in $\widehat{T}$.*
(ii) *If v is a leaf that is not youngest in T, then $\tilde{v}$ is a leaf in even level in $\widehat{T}$.*
(iii) *If v is a youngest leaf and there are k edges ($k > 0$) in the path from v to $\tilde{v}$ in T, then $\tilde{v}$ is an internal node with degree k in even-level in $\widehat{T}$. In particular,* $\mathsf{ystleaf}(T) = \mathsf{elint}(\widehat{T})$.

Moreover, $\mathsf{oleaf}(T) = \mathsf{oleaf}(\widehat{T})$.

Before proving Theorem 2.15, we need to introduce some definitions on trees. Let $B \in \mathcal{B}_M$ be a binary tree. A *right chain* (resp., *left chain*) of B is a maximal path composed of only right (resp., left) edges. The *leader* of a path in B is the node closest to the root in B. For instance, the second tree in Fig. 7 has five left chains, which are $1-6-7$, $8-10$, $2-3$, $4-5$ and 9 whose leaders are colored in magenta. A *hoe* of B is a path that begins with a right edge and continues with only left edges. For instance, $7-8-9$ is a hoe in the third tree in Fig. 7. Note that left/right chains and hoes in binary trees were first considered by Lin et al. [20]. For a tree $T \in \mathcal{T}_M$, a *claw* of T is a subtree formed by an internal node v together with all the children of v. For instance, the subtree induced by the nodes $7, 8$ and 9 in the last tree in Fig. 7 is a claw.

We are now ready to prove Theorem 2.15.

Proof of Theorem 2.15. (i) Since $\tilde{v}$ is the youngest child, it becomes a head in $\rho(T)$, which is the leader of its left chain whose members are its $q-1$ siblings. By Lemma 2.14, $\tilde{v}$ turns to be a node at odd right-level under θ. Thus, it becomes an odd-level node with degree $q-1$ in $\widehat{T}$.

(ii) In this case $v = \tilde{v}$, which is a leaf that is not youngest in T. Thus, under ρ it becomes a non-head without right child. By Lemma 2.14, $\tilde{v}$ turns to be a node without right child at even right-level under θ. Eventually, it turns to be a leaf at even-level in $\widehat{T}$ under ρ^{-1}.

(iii) Since $\tilde{v}$ is the first node that is not youngest in the path from v to the root in T, if letting P be the path from $\tilde{v}$ to v, then all the nodes in P are youngest, except $\tilde{v}$. Then under ρ, P becomes the right chain with $\tilde{v}$ as the leader. This right chain turns to a maximal hoe with $\tilde{v}$ as the leader under θ. Eventually, this hoe becomes a claw under ρ^{-1}, i.e., the node $\tilde{v}$ with all the remaining k nodes in the hoe as children in $\widehat{T}$. Thus, $\tilde{v}$ is of degree k and as $\tilde{v}$ is not youngest in T, $\tilde{v}$ lies at even level in $\widehat{T}$ according to Lemma 2.14.

Finally, by Lemma 2.2, old leaves in T become leaves in $\rho(T)$, which are preserved under θ. Thus, $\mathsf{oleaf}(T) = \mathsf{oleaf}(\widehat{T})$, as desired. □

One interesting consequence of the plane tree case of Theorem 2.15 is the following symmetry.

Corollary 2.16. *For $n \geq 1$, the pair* (oleaf, elint) *is symmetric on $\mathcal{P}_n$. In particular, the two statistics* oleaf *and* elint *are equidistributed on $\mathcal{P}_n$ and so*

$$|\{T \in \mathcal{P}_n : \mathsf{elint}(T) = k\}| = 2^{n-2k+1}\binom{n-1}{2k-2}C_{k-1}.$$

Proof. By Theorem 2.15, the pair (oleaf, elint) has the same joint distribution as the pair (oleaf, ystleaf) on $\mathcal{P}_n$. By the minor symmetry of plane trees, we see that (oleaf, ystleaf) is symmetric on $\mathcal{P}_n$, from which the symmetry (oleaf, elint) follows.

The second statement then follows from the fact

$$|\{T \in \mathcal{P}_n : \mathsf{oleaf}(T) = k\}| = 2^{n-2k+1}\binom{n-1}{2k-2}C_{k-1}, \tag{2.7}$$

which was proved in [4, Proposition 2]. Alternatively, formula (2.7) can also be deduced from (1.3) by setting $x_2 = y_1 = y_2 = 1$. □

Remark 2.17. By the minor symmetry of plane trees, the two triples (oleaf, ystleaf, elint) and (ystleaf, oleaf, elint) are equidistributed on $\mathcal{P}_n$. In particular, (ystleaf, elint) is symmetric on $\mathcal{P}_n$. By Theorem 2.15, the two statistics ystleaf and elint are equidistributed on $\mathcal{T}_M$. However, (ystleaf, elint) is not symmetric on $\mathcal{T}_n$, the set of all increasing trees on $[n]$.

Note also that oleaf and elint are not equidistributed on $\mathcal{T}_n$.

Another consequence of Theorem 2.15 for plane trees and the minor symmetry of plane trees is the following new interpretation for the refined Narayana polynomials.

Corollary 2.18. *For $n \geq 1$, we have the equidistribution*

$$\sum_{T\in\mathcal{P}_n} x^{\mathsf{oleaf}(T)}y^{\mathsf{yleaf}(T)} = \sum_{T\in\mathcal{P}_n} x^{\mathsf{elint}(T)}y^{\mathsf{elleaf}(T)}.$$

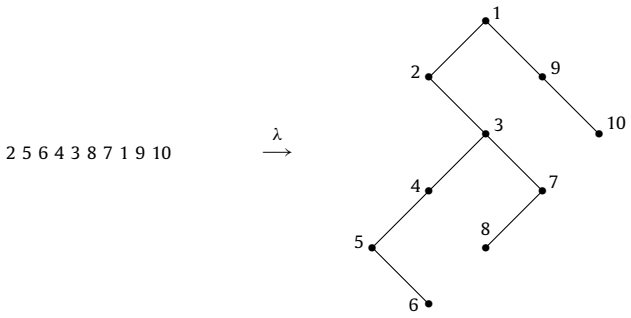


Fig. 8. An example of the bijection λ.

3. Applications to permutation patterns and statistics

In this section, we use a classical bijection between increasing binary trees $\mathcal{I}_n := \mathcal{B}_{[n]}$ and permutations $\mathcal{S}_n$ of $[n]$ to transform statistics from trees to permutations. In subsection 3.1, we consider two types of peak statistics on permutations corresponding to left and right leaves in binary increasing trees under λ. In particular, we prove a Riccati equation for the joint distribution of these two types of peaks on permutations. A special case of this Riccati equation is solved, which leads to a connection with the work of Elizalde and Noy [12] on consecutive patterns on permutations. Then in subsection 3.2, as applications of the two involutions ϕ and ψ on binary trees introduced in Section 2.2, we prove two symmetries in 312-avoiding permutations, one of which generalizes a result of Kitaev and Zhang [16].

Given a word $w = w_1 w_2 \cdots w_n$ of positive integers with no repeated letters, we can write it as $w = \sigma i \tau$ with i the least element of w. Set $\lambda(\emptyset) = \emptyset$ and define $\lambda(w)$ recursively as $(\lambda(\sigma), i, \lambda(\tau))$, the binary tree rooted at i with left branch $\lambda(\sigma)$ and right branch $\lambda(\tau)$. See Fig. 8 for an example of λ. The map $\lambda : \mathcal{S}_n \to \mathcal{I}_n$ is a bijection (see [25, Page 23]). The inverse map λ^{-1} can be viewed as projecting the nodes of the increasing binary trees downwards.

3.1. Refinements of the peak statistic on permutations

Given a permutation $\pi = \pi_1 \pi_2 \dots \pi_n \in \mathcal{S}_n$, a letter π_i $(1 \le i \le n)$ is called a *peak* (resp., *double descent, double ascent*) of π if $\pi_{i-1} < \pi_i > \pi_{i+1}$ (resp., $\pi_{i-1} > \pi_i > \pi_{i+1}$, $\pi_{i-1} < \pi_i < \pi_{i+1}$), where by convention $\pi_0 = \pi_{n+1} = 0$. We further refer to a peak π_i as of *type 1* if $\pi_{i-1} \le \pi_{i+1}$ and of *type 2* otherwise. Let $\mathsf{pk}(\pi)$ (resp., $\mathsf{pk}_1(\pi)$, $\mathsf{pk}_2(\pi)$, $\mathsf{dd}(\pi)$, $\mathsf{da}(\pi)$) be the number of peaks (resp., type 1 peaks, type 2 peaks, double descents, double ascents) of π.

Lemma 3.1. *Let $\pi \in \mathcal{S}_n$ and $B = \lambda(\pi)$. For $n \ge 2$ and $i \in [n]$, we have*

(i) *π_i is a peak of type 1 in π iff π_i is a left leaf in B;*
(ii) *π_i is a peak of type 2 in π iff π_i is a right leaf in B;*
(iii) *π_i is a double descent in π iff π_i has only left child in B;*
(iv) *π_i is a double ascent in π iff π_i has only right child in B.*

Proof. Properties (iii) and (iv) are known in [25, Page 24], as well as the fact that π_i is a peak in π iff π_i is a leaf in B. Assertions (i) and (ii) then follow from the observation that π_i is a left (resp., right) leaf in B implies $\pi_{i-1} < \pi_{i+1}$ (resp., $\pi_{i-1} > \pi_{i+1}$). □

It is well known (see [25, Page 22]) that the *Eulerian polynomial* $A_n(x)$ enumerates permutations in $\mathcal{S}_n$ by number of descents. Introduce the refinement of Eulerian polynomials for $n \ge 1$ by

$$A_n(x, y, z_1, z_2) = \sum_{\pi \in \mathcal{S}_n} x^{\mathsf{dd}(\pi)} y^{\mathsf{da}(\pi)} z_1^{\mathsf{pk}_1(\pi)} z_2^{\mathsf{pk}_2(\pi)}.$$

The first three values of $A_n(x, y, z_1, z_2)$ are

$$A_1(x, y, z_1, z_2) = z_1,$$
$$A_2(x, y, z_1, z_2) = xz_1 + yz_2,$$
$$A_3(x, y, z_1, z_2) = x(x + y)z_1 + y(x + y)z_2 + 2z_1z_2.$$

Comparing Lemma 3.1 with Lemma 2.2, we have

$$A_n(x,y,z_1,z_2)=\sum_{T\in\mathcal{T}_n} x^{\mathsf{yleaf}(T)}y^{\mathsf{yint}(T)}z_1^{\mathsf{eleaf}(T)}z_2^{\mathsf{sleaf}(T)}$$

for $n\geq 2$. We are interested in computing the generating function

$$A(x,y,z_1,z_2;t)=\sum_{n\geq 1}A_n(x,y,z_1,z_2)\frac{t^n}{n!}$$

for two reasons. First, Carlitz and Scoville [1] proved the nice generating function formula

$$A(x,y,z,z;t)=\frac{uv(e^{ut}-e^{vt})}{ue^{vt}-ve^{ut}}, \tag{3.1}$$

where $u+v=x+y$ and $uv=z$. Second, Dong et al. [11] has obtained explicit algebraic generating function (see Theorem 2.11 for a refinement) for plane trees by the quadruple of statistics (eleaf, sleaf, yleaf, yint). Unfortunately, we are unable to find any explicit formula for $A(x,y,z_1,z_2;t)$. Instead, we can prove that $A(x,y,z_1,z_2;t)$ satisfies a Riccati equation.

Theorem 3.2. *The generating function $A=A(x,y,z_1,z_2;t)$ satisfies the Riccati equation:*

$$\frac{\partial A}{\partial t}=A^2+((z_2-z_1)t+x+y)A+(z_2-z_1)yt+z_1. \tag{3.2}$$

Proof. For $1\leq i\leq n$, let $A_{n,i}=A_{n,i}(x,y,z_1,z_2):=\sum_{\pi\in\mathcal{S}_{n,i}}x^{\mathsf{dd}(\pi)}y^{\mathsf{da}(\pi)}z_1^{\mathsf{pk}_1(\pi)}z_2^{\mathsf{pk}_2(\pi)}$, where $\mathcal{S}_{n,i}$ is the set of permutations $\pi\in\mathcal{S}_n$ with $\pi_i=1$. For convenience, we write $A_n=A_n(x,y,z_1,z_2)$. For $n\geq 4$ and $\pi\in\mathcal{S}_{n,i}$, we write $\pi=\sigma 1\tau$ and consider the following five cases.

(1) If $i=1$, then we have

$$(\mathsf{dd},\mathsf{pk}_1,\mathsf{pk}_2)(\pi)=(\mathsf{dd},\mathsf{pk}_1,\mathsf{pk}_2)(\tau)\text{ and }\mathsf{da}(\pi)=1+\mathsf{da}(\tau).$$

Thus, $A_{n,1}=yA_{n-1}$.

(2) If $i=n$, then we have

$$(\mathsf{da},\mathsf{pk}_1,\mathsf{pk}_2)(\pi)=(\mathsf{da},\mathsf{pk}_1,\mathsf{pk}_2)(\sigma)\text{ and }\mathsf{dd}(\pi)=1+\mathsf{dd}(\sigma).$$

Thus, $A_{n,n}=xA_{n-1}$.

(3) If $i=2$, then we have

$$(\mathsf{dd},\mathsf{da},\mathsf{pk}_2)(\pi)=(\mathsf{dd},\mathsf{da},\mathsf{pk}_2)(\tau)\text{ and }\mathsf{pk}_1(\pi)=1+\mathsf{pk}_1(\tau).$$

Thus, $A_{n,2}=(n-1)z_1A_{n-2}$.

(4) If $i=n-2$, then we have

$$(\mathsf{dd},\mathsf{da},\mathsf{pk}_1)(\pi)=(\mathsf{dd},\mathsf{da},\mathsf{pk}_1)(\sigma)\text{ and }\mathsf{pk}_2(\pi)=1+\mathsf{pk}_2(\sigma).$$

Thus, $A_{n,2}=(n-1)z_2A_{n-2}$.

(5) If $3\leq i\leq n-2$, then we have

$$(\mathsf{dd},\mathsf{da},\mathsf{pk}_1,\mathsf{pk}_2)(\pi)=(\mathsf{dd},\mathsf{da},\mathsf{pk}_1,\mathsf{pk}_2)(\sigma)+(\mathsf{dd},\mathsf{da},\mathsf{pk}_1,\mathsf{pk}_2)(\tau).$$

Thus, $A_{n,i}=\binom{n-1}{i-1}A_{i-1}A_{n-i}$ for $3\leq i\leq n-2$.

Summing over all cases gives

$$A_n=(x+y)A_{n-1}+(n-1)(z_1+z_2)A_{n-2}+\sum_{i=2}^{n-3}\binom{n-1}{i}A_iA_{n-1-i} \tag{3.3}$$

for $n\geq 4$. Involving the first three values of $A_n(x,y,z_1,z_2)$, we can turn this recurrence relation into the Riccati equation (3.2). □

Remark 3.3. A grammatical calculus approach to Carlitz-Scoville's formula (3.1) was given by Fu [13] and it would be interesting to see whether her approach could be adopted to the refinement $A(x,y,z_1,z_2;t)$.

We could not solve the Riccati equation (3.2) to get explicit expression for $A(x,y,z_1,z_2;t)$. However, the special case that $x=y=z_2=1$ can be solved.

Proposition 3.4. *Let* $A(z;t)=\sum_{n\geq 1}\sum_{\pi\in\mathcal{S}_n} z^{\mathsf{pk}_1(\pi)}\frac{t^n}{n!}$. *Then,*

$$A(z;t)=\frac{e^{(z-1)t^2/2}}{1-\int_0^t e^{(z-1)x^2/2}dx}+(z-1)t-1. \tag{3.4}$$

Proof. By setting $x=y=z_2=1$ and $z_1=z$ in (3.2), we get the Riccati equation for $A(z;t)$

$$\frac{\partial A(z;t)}{\partial t}=A(z;t)^2+((1-z)t+2)A(z;t)+(1-z)t+z. \tag{3.5}$$

We observe that $(z-1)t-1$ is a particular solution of (3.5). Setting $B(z;t)=A(z;t)-(z-1)t+1$, we get the Bernoulli differential equation:

$$\frac{\partial B(z;t)}{\partial t}+(1-z)tB(z;t)=B(z;t)^2.$$

Letting $C(z;t)=B(z;t)^{-1}$, we obtain the following linear first-order ordinary differential equation:

$$\frac{\partial C(z;t)}{\partial t}+(z-1)tC(z;t)+1=0.$$

By solving this differential equation, we can express $C(z;t)$ as:

$$C(z;t)=c_1e^{(1-z)t^2/2}-c_2e^{(1-z)t^2/2}\int e^{(z-1)t^2/2}dt,$$

where c_1 is an arbitrary constant and c_2 is an arbitrary positive constant. Therefore, we can write:

$$A(z;t)=\left(c_1e^{(1-z)t^2/2}-c_2e^{(1-z)t^2/2}\int e^{(z-1)t^2/2}dt\right)^{-1}+(z-1)t-1.$$

Involving $A(z;0)=0$ and $A'(z;0)=z$, we get (3.4). □

For $\sigma\in\mathcal{S}_k$ and a word $w=w_1w_2\cdots w_n$ of positive integers, an occurrence of (consecutive pattern) σ in w is a subword $w_iw_{i+1}\cdots w_{i+k-1}$ that is order isomorphic to σ. Let $\underline{\sigma}(w)$ denote the number of occurrences of σ in w. Then, $\mathsf{pk}_1(\pi)=\underline{132}(0\pi)$ and $\mathsf{pk}_2(\pi)=\underline{231}(\pi 0)$ for any $\pi\in\mathcal{S}_n$ with $n\geq 2$. Consider the enumerator of permutations by the number of occurrences of 132 as

$$B_n(z):=\sum_{\pi\in\mathcal{S}_n} z^{\underline{132}(\pi)}.$$

Elizalde and Noy [12, Theorem 4.1] proved that

$$\sum_{n\geq 0}B_n(z)\frac{t^n}{n!}=\frac{1}{1-\int_0^t e^{(z-1)x^2/2}dx},$$

where by convention $B_0(z)=1$. Comparing with (3.4) we get the following relationship.

Corollary 3.5. *For* $n\geq 2$, *we have*

$$\sum_{\pi\in\mathcal{S}_n} z^{\mathsf{pk}_1(\pi)}=\sum_{k=0}^{\lfloor n/2\rfloor}\frac{(z-1)^k}{2^k}\binom{n}{k}B_{n-2k}(z).$$

It would be interesting to find a combinatorial proof of the above relationship.

3.2. Symmetries in 312-avoiding permutations

A permutation $\pi\in\mathcal{S}_n$ is said to be 312-*avoiding* if there are no indices $1\leq i<j<k\leq n$ such that $\pi_i>\pi_k>\pi_j$. If we denote $\mathcal{S}_n(312)$ the set of all 312-avoiding permutations in $\mathcal{S}_n$, then a classical result of Knuth (see [14, Section 2.1.1]) asserts that $|\mathcal{S}_n(312)|=C_n$. In this subsection, we prove two symmetries in 312-avoiding permutations involving maximum number of non-overlapping ascents/descents and some consecutive patterns of length 4.

Let us recall some classical permutation statistics. An *ascent* (resp., A *descent*) in a permutation $\pi\in\mathcal{S}_n$ is an index $i\in[n-1]$ such that $\pi_i<\pi_{i+1}$ (resp., $\pi_i>\pi_{i+1}$). Denote by $\mathrm{ASC}(\pi)$ and $\mathrm{DES}(\pi)$ the set of all ascents and the set of all descents of π, respectively. Let $\mathsf{asc}(\pi):=|\mathrm{ASC}(\pi)|$ and $\mathsf{des}(\pi):=|\mathrm{DES}(\pi)|$. Furthermore, the number of *left-to-right*

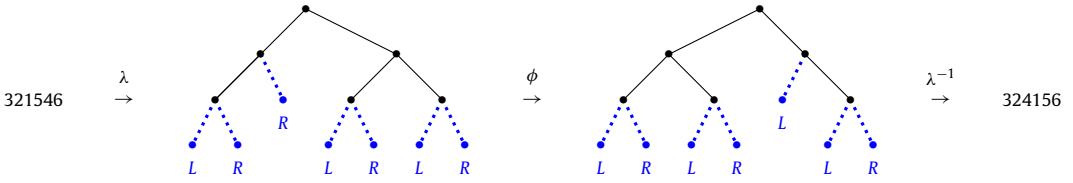


Fig. 9. An example of the involution Λ.

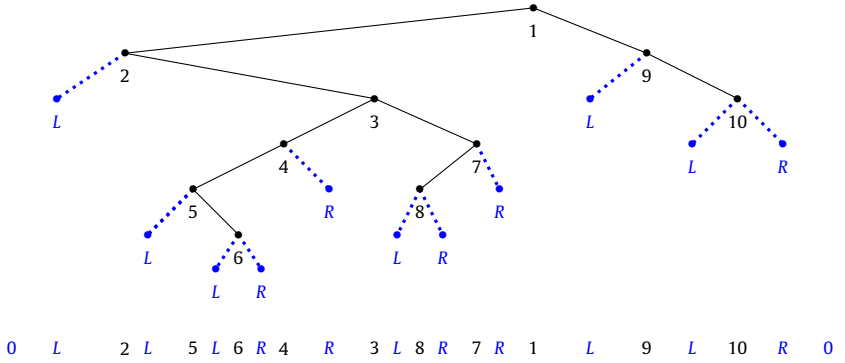


Fig. 10. An example of the complement of a binary tree $\bar{B}$.

(resp., *right-to-left*) *minima* of π is denoted by $\mathsf{lmin}(\pi)$ (resp., $\mathsf{rmin}(\pi)$). The *left descending run* (resp., *right ascending run*) of π, namely the largest $j \in [n]$ such that the subsequence $\pi_1\pi_2\cdots\pi_j$ (resp., $\pi_{n-j+1}\pi_{n-j+2}\ldots\pi_n$) of π is decreasing (resp., increasing), is denoted as $\mathsf{ldr}(\pi)$ (resp., $\mathsf{rar}(\pi)$).

Two indices i and j, $i < j$, are said to be *overlapping* if $j = i+1$. Kitaev [15] introduced the *maximum number of non-overlapping ascents* (resp., *descents*) in a permutation π, denoted $\mathsf{mna}(\pi)$ (resp., $\mathsf{mnd}(\pi)$). For example, $\mathsf{mna}(231485697) = 3$ and $\mathsf{mnd}(126543) = 2$. Recently, Kitaev and Zhang [16] investigated the statistics mna and mnd over stack-sortable permutations (i.e., 231-avoiding permutations) and proved a symmetry that is equivalent to the following result.

Theorem 3.6 *(Kitaev-Zhang [16]). The following two* 8*-tuples of statistics*

$$(\mathsf{mnd}, \mathsf{mna}, \mathsf{des}, \mathsf{asc}, \mathsf{ldr}, \mathsf{rar}, \mathsf{lmin}, \mathsf{rmin})$$

$$(\mathsf{mna}, \mathsf{mnd}, \mathsf{asc}, \mathsf{des}, \mathsf{rar}, \mathsf{ldr}, \mathsf{rmin}, \mathsf{lmin}),$$

are equidistributed over $\mathcal{S}_n(312)$ *for* $n \geq 1$.

As an application of the minor symmetry of binary trees, we can generalize the above result as follows.

Theorem 3.7. *For* $\pi \in \mathcal{S}_n$, *let* $\widetilde{\mathsf{ASC}}(\pi) := \{n-i : i \in \mathsf{ASC}(\pi)\}$. *There exists an involution* Λ *on* $\mathcal{S}_n(312)$ *that transforms the triple* $(\mathsf{DES}, \mathsf{pk}_1, \mathsf{lmin})$ *to* $(\widetilde{\mathsf{ASC}}, \mathsf{pk}_2, \mathsf{rmin})$. *In particular,* Λ *provides an involution proof of Theorem 3.6.*

Proof. For $\pi \in \mathcal{S}_n(312)$, it is plain to see that the nodes of the increasing binary tree $\lambda(\pi)$ are exactly labeled according to their preorder (see Fig. 8 where the permutation is 312-avoiding). Thus, we can remove all labels of $\lambda(\pi)$ and denote the resulting binary tree by $\tilde{\lambda}(\pi)$. It is clear that $\tilde{\lambda} : \mathcal{S}_n(312) \to \mathcal{B}_n$ is a bijection. Now define $\Lambda = \tilde{\lambda}^{-1} \circ \phi \circ \tilde{\lambda}$, where ϕ is the minor symmetry on $\mathcal{B}_n$. See Fig. 9 for an example of Λ.

Let $B = \lambda(\pi)$. To facilitate the proof, we introduce the *complement of a binary tree*. We assign leaves labeled by L or R to B to obtain a complete binary tree, ensuring that all original nodes in B become internal nodes. We use $\bar{B}$ to denote such a complement of B where the added leaf in $\bar{B}$ is labeled by L iff it is a left leaf and R iff it is a right leaf. See Fig. 10 for an example of $\bar{B}$. It is plain to see that the index i is a descent of π iff the $(i+1)$-th (under preorder) new leaf in $\bar{B}$ is R. Thus, $\mathsf{DES}(\pi) = \widetilde{\mathsf{ASC}}(\Lambda(\pi))$.

The fact that $\mathsf{pk}_1(\pi) = \mathsf{pk}_2(\Lambda(\pi))$ follows from Lemma 3.1 and the construction of Λ.

The *left arm* (resp., *right arm*) of B is the maximal chain starting from the root and compositing only left (resp., right) edges. From the inverse map λ^{-1}, it is clear that $\mathsf{lmin}(\pi)$ (resp., $\mathsf{rmin}(\pi)$) equals the number of nodes in the left (resp., right) arm of B. The fact that $\mathsf{lmin}(\pi) = \mathsf{rmin}(\Lambda(\pi))$ then follows from the construction of Λ.

Finally, as $\mathsf{DES}(\pi)$ (or $\widetilde{\mathsf{ASC}}(\pi)$) determines the sextuple $(\mathsf{mnd}, \mathsf{mna}, \mathsf{des}, \mathsf{asc}, \mathsf{ldr}, \mathsf{rar})(\pi)$, we see that Λ provides an involution proof of Theorem 3.6. □

For a permutation π, let $\mathrm{st}_1(\pi) := \underline{1324}(0\pi)$ and $\mathrm{st}_2(\pi) := \underline{3241}(\pi 0)$.

Theorem 3.8. *There exists an involution Υ on $\mathcal{S}_n(312)$ that preserves the triple* (pk, dd, da) *but exchanges the pair* $(\mathrm{st}_1, \mathrm{st}_2)$.

Proof. Recall the bijection $\tilde{\lambda} : \mathcal{S}_n(312) \to \mathcal{B}_n$ in the proof of Theorem 3.7 and the involution ψ on $\mathcal{B}_n$ introduced in the proof of Theorem 1.9. Set $\Upsilon = \tilde{\lambda}^{-1} \circ \psi \circ \tilde{\lambda}$. It follows from Lemma 3.1 and the construction of ψ that Υ preserves the triple (pk, dd, da).

To demonstrate that Υ exchanges the pair $(\mathrm{st}_1, \mathrm{st}_2)$, we claim that each consecutive 1324 (resp., 3241) pattern in $\pi \in \mathcal{S}_n(312)$ corresponds to a left (resp., right) leaf whose parent has right (resp., left) child in the binary tree $B = \lambda(\pi)$. If π contains a consecutive 1324 pattern $\pi_{i-1}\pi_i\pi_{i+1}\pi_{i+2}$, then π_i is a peak of type 1 and so π_i is a left leaf by Lemma 3.1 (i). Moreover, $\pi_{i+2} > \pi_{i+1}$ implies that π_{i+1} has a right child (which is not necessarily π_{i+2}). Conversely, if the node π_i is a left leaf in B whose parent has right child, then $\pi_{i-1}\pi_i\pi_{i+1}$ is a peak of type 1 by Lemma 3.1 (i). Furthermore, as π_{i+1} is the parent of π_i in B, it has right child. Thus, $\pi_{i-1} < \pi_{i+1} < \pi_i < \pi_{i+2}$ and so $\pi_{i-1}\pi_i\pi_{i+1}\pi_{i+2}$ forms a consecutive 1324 pattern in π. This proves that $\mathrm{st}_1(\pi) = \mathsf{entleaf}(B)$. The fact that $\mathrm{st}_2(\pi) = \mathsf{suleaf}(B)$ follows by similar discussions. As ψ exchanges the pair (entleaf, suleaf) on $\mathcal{B}_n$, Υ exchanges the pair $(\mathrm{st}_1, \mathrm{st}_2)$ on $\mathcal{S}_n(312)$. This completes the proof of the theorem. □

Declaration of competing interest

The authors confirm that there is not any conflict of interest caused by this paper.

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Data availability

No data was used for the research described in the article.

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